

Qiang Fu

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Experience

2026–present	Senior Research Scientist	Center of Excellence for Generative AI, King Abdullah University of Science and Technology (KAUST), Saudi Arabia
2017–2026	Research Scientist	Visual Computing Center, KAUST, Saudi Arabia
2016–2017	Research Associate Professor	School of Information Science and Technology, ShanghaiTech University, China
2014–2016	Postdoctoral Research Fellow	Visual Computing Center, KAUST, Saudi Arabia
2012–2014	Assistant Research Fellow	Key Laboratory of Computational Optical Imaging Technology, Academy of Opto-Electronics, Chinese Academy of Sciences, China

Education

2007–2012	Ph.D., Optical Engineering	University of Chinese Academy of Sciences (UCAS), China
2003–2007	B.Sc., Mechanical Engineering	University of Science and Technology of China (USTC), China

Research

My research centers on **computational imaging**, where optical hardware and computational algorithms are jointly designed to shape how visual information is acquired, encoded, reconstructed, and interpreted. Rather than following the conventional sequential pipeline from optical design to downstream image processing, I treat optics and computation as parts of a coherent system design problem. This allows acquisition and optical encoding at the front end to be co-designed with reconstruction and interpretation at the computational back end. Recent advances in AI, particularly differentiable and learning-based methods, make this system-level joint design increasingly tractable. These methods enable joint optimization of acquisition, encoding, reconstruction, and interpretation within a physics-grounded framework. Diffractive optical elements (DOEs) provide a powerful physical platform for this joint design, enabling engineered optical encoding to be embedded directly into the imaging front end. My long-standing experience in micro/nanofabrication enables me to carry such co-designed systems from computational models through fabricated components and integrated prototypes to experimental validation. This design-to-fabrication workflow has enabled computational sensing and imaging across a range of optical information dimensions and system capabilities, including hyperspectral and polarization imaging, high-dynamic-range imaging, wavefront sensing, and adaptive optics. Overall, my research takes a system-level approach to computational imaging, spanning imaging concepts and computational methods, optical design, fabrication, and prototype realization, toward next-generation computational cameras and imaging systems in which sensing and computation are co-designed from the outset to enable new imaging capabilities and translation into practical imaging technologies.

Expertise

- **Computational methods:** Image formation and inverse problems; image signal processing; differentiable simulation and optimization; physics-grounded deep learning.
- **Optical system design:** Refractive, diffractive, and hybrid optical systems; wave-optics; ray-tracing simulation; optical encoding; optomechanical design, prototyping, and optical metrology.
- **Micro/nanofabrication:** Design and fabrication of diffractive optical elements and metasurfaces; cleanroom process development spanning photolithography, laser direct writing, e-beam lithography, two-photon polymerization, reactive-ion etching, thin-film deposition, nanoimprinting; parametric GDSII layout and prototype integration.

Teaching and Supervision

- **Research supervision and mentoring at KAUST:** Mentored and co-supervised 13 master's and doctoral students and 4 postdoctoral researchers in Prof. Wolfgang Heidrich's research group at KAUST, 2017–present. Provided guidance on research conceptualization, system design, prototyping, experimentation, publication, and career development.
- **Research mentoring at ShanghaiTech:** Mentored 3 master's and doctoral students in Prof. Jingyi Yu's research group at the School of Information Science and Technology, ShanghaiTech University, 2016–2017.
- **Teaching certificate:** Completed the Harvard Bok Higher Education Teaching Certificate online short course, 2023.

- **Guest teaching:** Delivered remote guest lectures on computational diffractive imaging in Prof. Jingang Zhang's course at the University of Chinese Academy of Sciences, 2020.
- **Outreach:** Supervised a student art-science project on metamerism through The KAUST School Action Service Program and participated in science outreach and Visual Computing Center Open House activities at KAUST.

Academic Service

- **Associate Editor**, *Optics Continuum*, 2026–present.
- **Technical Program Committee Member**, ACM SIGGRAPH Asia, Hong Kong (2025) and Kuala Lumpur (2026).
- **Poster Co-Chair**, IEEE International Conference on Computational Photography (ICCP), Princeton, 2026.
- **Member**, CEMSE Divisional Internal Review Committee for the KAUST Global Fellowship, 2026.
- **Outstanding Reviewer Recognition**, Optica Publishing Group, 2024.

Journal reviewing: Springer Nature: *Nature*, *Nature Communications*, *Communications Engineering*, *International Journal of Computer Vision*, *Computational Visual Media*, and *Light: Advanced Manufacturing*; ACM: *ACM Transactions on Graphics* and *ACM Transactions on Design Automation of Electronic Systems*; IEEE: *IEEE Transactions on Image Processing*, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Visualization and Computer Graphics*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Computational Imaging*, *IEEE Transactions on Industrial Informatics*, *IEEE/ASME Transactions on Mechatronics*, *IEEE Transactions on Circuits and Systems for Video Technology*, and *IEEE Transactions on Neural Networks and Learning Systems*; Elsevier: *Computers & Graphics*, *Optics and Lasers in Engineering*, *Measurement*, *Expert Systems with Applications*, *Neurocomputing*, *Optics & Laser Technology*, *Data in Brief*, *Biomedical Signal Processing and Control*, *Engineering Applications of Artificial Intelligence*, *Information Processing in Agriculture*, *Medical Image Analysis*, and *Materials Science in Semiconductor Processing*; Optica Publishing Group: *Optica*, *Optics Express*, *Optics Letters*, *Applied Optics*, *Optics Continuum*, *Journal of the Optical Society of America A*, *Optical Materials Express*, *Photonics Research*, and *Biomedical Optics Express*; Wiley: *Laser & Photonics Reviews* and *Journal of Field Robotics*; IOP Publishing: *Inverse Problems*.

Conference reviewing: ACM SIGGRAPH (2016, 2019, 2022, 2025, 2026); ACM SIGGRAPH Asia (2017, 2023, 2024, 2025, 2026); ACM Multimedia (2025); IEEE/CVF CVPR (2024, 2025, 2026); IEEE/CVF ICCV (2025); ECCV (2026); and AAAI (2025).

Research Recognition

- **Year in Optics 2024**, *Optics & Photonics News*: “Curriculum learning for *ab initio* deep learned refractive optics.”
- **Year in Optics 2024**, *Optics & Photonics News*: “Neural Étendue expander for ultra-wide-angle high-fidelity holographic display.”
- **KAUST Discovery Feature**: “Deep learning benefits lens design,” 2024.
- **Top Downloaded Article, 2021–2022**, *Wiley Computer Graphics Forum*: “Linear polarization demosaicking for monochrome and colour polarization focal plane arrays.”
- **Best Poster Award**: KAUST Computational Imaging and Vision Conference, 2017.

Talks and Presentations

Invited Talks and Seminars

- **Diffractive Optical Elements for Computational Imaging: From Theory to Fabrication.** Princeton University, Princeton, USA, April 2026 (virtual).
- **Limitations of Data-Driven Spectral Reconstruction: An Optics-Aware Analysis.** Joint HKU-KAUST Workshop, Hong Kong, China, December 2025.
- **Compact Computational Imaging: Diffractive Lens, Metalens, and Hybrid Lens.** Xiaomi Inc., Beijing, China, June 2024.
- **Computational Imaging with Diffractive Optics.** Beijing Academy of Artificial Intelligence, November 2021 (virtual).
- **Applications of Diffractive Optical Elements (DOEs) in High-Dynamic-Range Computational Imaging.** International Computational Imaging Conference (CITA2021), Hangzhou, China, September 2021 (virtual).
- **Computational Diffractive Imaging: Where Computation and Optics Meet.** Northwestern Polytechnical University, Xi'an, China, April 2017.
- **Computational Imaging: A Systematic View.** Institute of Automation, Chinese Academy of Sciences, Beijing, China, May 2016.
- **Computational Diffractive Imaging: Design, Fabrication and Applications.** Dagstuhl Seminar on Computational Imaging, Dagstuhl, Germany, May 2015.
- **Diffractive Optical Elements: Theories, Methods, Fabrication and Applications in Computational Imaging.** Academy of Opto-Electronics, Chinese Academy of Sciences, Beijing, China, March 2015.

Conference Presentations

- **End-to-End Hybrid Refractive-Diffractive Lens Design with Differentiable Ray-Wave Model.** ACM SIGGRAPH Asia, Tokyo, Japan, December 2024.
- **Limitations of Data-Driven Spectral Reconstruction: An Optics Perspective.** Optica Imaging Congress, Toulouse, France, July 2024.
- **Megapixel Adaptive Optics: Towards Correcting Large-Scale Distortions in Computational Cameras.** ACM SIGGRAPH, Vancouver, Canada, August 2018. (Presented with Congli Wang)

Books

- [1] Jingang Zhang, Yunfeng Nie, **Qiang Fu**, and Xiongzhi Wang. "Computational Optical Engineering – A Python Tutorial." *Xidian University Press*, Xi'an, 2023. In Chinese. ISBN: 978-7-5606-6708-9.

Patents

- [1] Xinge Yang, **Qiang Fu**, and Wolfgang Heidrich. "Method for fully automatic lens design with curriculum learning." *U.S. Provisional Patent Application No. 63/358,583*, filed July 2022.
- [2] Ilya Chugunov, Seung-Hwan Baek, **Qiang Fu**, Wolfgang Heidrich, and Felix Heide. "Microlens amplitude masks for flying pixel removal in time-of-flight imaging." *International Patent Publication WO 2022/195537 A1*, 2022.
- [3] Congli Wang, Xiong Dun, **Qiang Fu**, and Wolfgang Heidrich. "Wavefront sensor and method of reconstructing distorted wavefronts." *International Patent Publication WO 2018/083573 A1*, May 11, 2018.
- [4] Bin Xiangli, **Qiang Fu**, and Qunbo Lü. "A lateral shearing interferometer with dual-channel outputs." *Chinese Patent Application No. CN102519609A*.

Publications

- [1] Yidan Zheng, Matheus Souza, Kaizhang Kang, **Qiang Fu**, Hadi Amata, Wolfgang Heidrich. "Low Latency Gaze Tracking via Latent Optical Sensing." *ACM SIGGRAPH Asia*, 2026. Accepted.
- [2] Baoyuan Zhang, Shiyao Guo, Lin Tang, Yi Chen, Lukas Glandorf, Etienne Jessen, Xuyang Chang, Tian Jin, Michael Reiss, Shuxin Lyu, **Qiang Fu**, Hadi Amata, Wolfgang Heidrich, Chaim Glück, Dominik Schillinger, Bruno Weber, Xosé Luís Deán-Ben, Weibo Wang, Xiong Dun, Daniel Razansky, Zhenyue Chen, and Quanyu Zhou. "Double-helix optical point spread function enables real-time mesoscopic 3D functional microangiography in the living mouse brain and skull." *Nat. Commun.*, 17, 5167, 2026.
- [3] Xinge Yang, **Qiang Fu**, Yunfeng Nie, and Wolfgang Heidrich. "Task-driven lens design." *Opt. Express*, 34(5): 8961–8975, 2026.
- [4] Juan Romero, **Qiang Fu**, Matteo Ravasi, and Wolfgang Heidrich. "Uncertainty quantification in HSI reconstruction using physics-aware diffusion priors and optics-encoded measurements." *arXiv preprint arXiv:2511.18473*, 2025.
- [5] Hadi Amata, **Qiang Fu**, and Wolfgang Heidrich. "Scalable and precise diffractive optics via grayscale lithography and soft nanoimprinting." *SPIE OptiFab*, 13728, pp.175–181, 2025.
- [6] Kaixuan Wei, Hector A. Jimenez-Romero, Hadi Amata, Jipeng Sun, **Qiang Fu**, Felix Heide, and Wolfgang Heidrich. "Large-area fabrication-aware computational diffractive optics." *ACM Trans. Graph. (Proc. SIGGRAPH Asia)*, 44(6), 243, 2025.
- [7] Jinhui Xiang, Yuqi Li, Jiabao Li, Wenxing Zheng, and **Qiang Fu**. "Self-calibrating fisheye lens aberrations for novel view synthesis." *Comput. Graph. Forum*, 44(6), e70148, 2025.
- [8] **Qiang Fu**, Matheus Souza, Eunsue Choi, Suhyun Shin, Seung-Hwan Baek, and Wolfgang Heidrich. "Limitations of data-driven spectral reconstruction: an optics-aware analysis." *IEEE Trans. Comput. Imaging*, 11, 1505–1520, 2025.
- [9] Haoyu Wei, Xin Liu, Yuhui Liu, **Qiang Fu**, Wolfgang Heidrich, Edmund Y. Lam, and Yifan Peng. "Learned off-aperture encoding for wide field-of-view RGBD imaging." *IEEE Trans. Pattern Anal. Mach. Intell.*, (Proc. ICCP) 2025.
- [10] Matheus Souza, Yidan Zheng, Kaizhang Kang, Yogeshwar Nath Mishra, **Qiang Fu**, and Wolfgang Heidrich. "Latent Space Imaging." *IEEE/CVF CVPR*, pp.28295–28305, 2025.
- [11] Yuhui Liu, Liangxun Ou, **Qiang Fu**, Hadi Amata, Wolfgang Heidrich, and Yifan Peng. "Learned binocular-encoding optics for RGBD imaging using joint stereo and focus cues." *IEEE/CVF CVPR*, pp.15833–15842, 2025.
- [12] Xinge Yang, **Qiang Fu**, Mohamed Elhoseiny, and Wolfgang Heidrich. "Aberration-aware depth-from-focus." *IEEE Trans. Pattern Anal. Mach. Intell.*, 47(9): 7268–7278, 2025.
- [13] Xinge Yang, Matheus Souza, Kunyi Wang, Praneeth Chakravarthula, **Qiang Fu**, and Wolfgang Heidrich. "End-to-end hybrid refractive-diffractive lens design with differentiable ray-wave model." *ACM Proc. SIGGRAPH Asia*, 47, pp. 1–11, 2024.
- [14] Zhisheng Zhou, Jingang Zhang, **Qiang Fu**, and Yunfeng Nie. "Linearized wavefront sensing model for aberration retrieval from low-frequency Fourier coefficients." *J. Opt. Soc. Am. A*, 41(11):C55–C61, 2024.
- [15] Christian Marois, Olivier Lardi re, Joeleff Fitzsimmons, Mathieu Demers, Adam B. Johnson, Saugata Dutt, Patrick Nkwari, Benjamin L. Gerard, Jennifer S. Dunn, Duan Li, **Qiang Fu**, Denis Brousseau, Suresh Sivanandam, William R.

- Thompson, Christopher Mann, Garima Singh, Marc-André Boucher, Tim Hardy, Krzysztof Caputa, Simon Thibault, Dmitry Savransky, Wolfgang Heidrich, Maaiké A. M. van Kooten, Jean-Pierre Véran, Rebecca Jensen-Clem, Darryl Gamroth, Colin Bradley, Sanjana Rawat, Öykü S. Galvan, Glen Herriot, and Jean-Thomas Landry. "CAL2: project update of the NRC Canada facility-class focal plane wavefront sensor for the Gemini Planet Imager 2 upgrade." *SPIE Adaptive Optics Systems IX*, 13097, 1309708, 2024.
- [16] Jiaqi Guo, Yujie Xing, Hadi Amata, **Qiang Fu**, Zhanshan Wang, Xiong Dun, and Xinbin Cheng. "Learned B-spline parametrization of lattice focal coding for monocular RGBD imaging." *IEEE International Conference on Computational Photography (ICCP)*, pp.1–12, 2024.
- [17] **Qiang Fu**, Matheus Souza, Eunsue Choi, Suhyun Shin, Seung-Hwan Baek, and Wolfgang Heidrich. "Limitations of hyperspectral imaging from RGB images: a data perspective." *Optica Imaging Congress (Computational Optical Sensing and Imaging)*, CTh4B.3, 2024.
- [18] Xiaoyin Mei, Yuqi Li, **Qiang Fu**, and Wolfgang Heidrich. "Progressive self-supervised learning for CASSI computational spectral cameras." *IEEE Trans. Comput. Imaging*, 10, pp.1505–1518, 2024.
- [19] Hadi Amata, **Qiang Fu**, and Wolfgang Heidrich. "Comparative performance analysis of multi-level diffractive lens and lens fabricated by grayscale lithography and soft-imprinting." *Optica Imaging Congress (Imaging Systems and Applications)*, 2024.
- [20] Xinge Yang, **Qiang Fu**, and Wolfgang Heidrich. "Curriculum learning for *ab initio* deep learned refractive optics." *Nat. Commun.*, 15, 6572, 2024.
- [21] Zheng Shi, Ilya Chugunov, Mario Bijelic, Geoffroi Côté, Jiwoon Yeom, **Qiang Fu**, Hadi Amata, Wolfgang Heidrich, and Felix Heide. "Split-aperture 2-in-1 computational cameras." *ACM Trans. Graph. (Proc. SIGGRAPH)*, 43(4), 141, 2024.
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- [23] Yidan Zheng, **Qiang Fu**, Hadi Amata, Praneeth Chakravarthula, Felix Heide, and Wolfgang Heidrich. "Hexagonal diffractive optical Elements." *Opt. Express*, 31(26): 43864–43876, 2023.
- [24] Hadi Amata, **Qiang Fu**, and Wolfgang Heidrich. "Additive fabrication of SiO_2 -based micro-optics with lag-free depth and reduced roughness." *Opt. Express*, 31(25): 41533–41545, 2023.
- [25] Hadi Amata, **Qiang Fu**, and Wolfgang Heidrich. "Additive diffractive optical elements fabrication by PECVD deposition of SiO_2 and lift-off process." *SPIE Digital Optical Technologies*, 12624, pp. 23–28, 2023.
- [26] Qizhou Wang, **Qiang Fu**, Maksim Makarenko, Arturo Burguete Lopez, Wolfgang Heidrich, and Andrea Fratalocchi. "Data-driven broadband achromatic metalens via first-principle end-to-end inverse design." *Optica CLEO: Applications and Technology*, JW2A.93, 2023.
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- [28] Yuqi Li, **Qiang Fu**, and Wolfgang Heidrich. "Extended depth-of-field projector using learned diffractive optics." *IEEE Virtual Reality*, pp. 449–459, 2023.
- [29] **Qiang Fu**, Dong-Ming Yan, and Wolfgang Heidrich. "Diffractive lensless imaging with optimized Voronoi-Fresnel phase." *Opt. Express*, 30(25): 45807–45823, 2022.
- [30] Zhisheng Zhou, Jingang Zhang, **Qiang Fu**, and Yunfeng Nie. "Phase-diversity wavefront sensing enhanced by a Fourier-based neural network." *Opt. Express*, 30(19): 34396–34410, 2022.
- [31] William Thompson, Christian Marois, Garima Singh, Olivier Lardi re, Benjamin Gerard, **Qiang Fu**, and Wolfgang Heidrich. "Performance of the FAST self coherent camera at the NEW-EARTH lab and a simplified SCC measurement algorithm." *SPIE Adaptive Optics Systems VIII*, vol. 12185, pp. 724–735, 2022.
- [32] Garima Singh, William Thompson, Olivier Lardi re, Christian Marois, Mamadou N'Diaye, Adam B Johnson, Jean-Pierre Véran, Glen Herriot, Benjamin Gerard, **Qiang Fu**, and Wolfgang Heidrich. "Pupil-plane LLOWFS simulation and laboratory results from NEW-EARTH's high-contrast imaging testbed." *SPIE Adaptive Optics Systems VIII*, vol. 12185, pp. 1592–1603, 2022.
- [33] Olivier Lardi re, Christian Marois, William R. Thompson, Garima Singh, Adam Johnson, Joeleff Fitzsimmons, Tim Hardy, Mamadou N'Diaye, Wolfgang Heidrich, **Qiang Fu**, Denis Brousseau, and Simon Thibault. "Optical design of SPIDERS, a Subaru Pathfinder Instrument for Detecting Exoplanets and Retrieving Spectra." *SPIE Adaptive Optics Systems VIII*, 12185, 1218547, 2022.
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- [38] Xinge Yang, **Qiang Fu**, and Wolfgang Heidrich. "Automatic lens design based on differentiable ray-tracing." *Optica Computational Optical Sensing and Imaging*, pp. CTh4C–2, 2022.
- [39] Xiong Dun, **Qiang Fu**, Haotian Li, Tiancheng Sun, Jian Wang, and Qilin Sun. "Recent progress in computational imaging." *J. Image Graph.*, 27(6): 1840–1876, 2022.
- [40] Youxin Pang, Mengke Yuan, **Qiang Fu**, Peiran Ren, and Dong-Ming Yan. "Progressive polarization based reflection removal via realistic training data generation." *Pattern Recognit.*, 124:108497, 2022.
- [41] **Qiang Fu**, Hadi Amata, Benjamin Gerard, Christian Marois, and Wolfgang Heidrich. "Additive lithographic fabrication of a Tilt-Gaussian-Vortex mask for focal plane wavefront sensing." *SPIE OptiFab 2021*, vol. 11889, pp. 162–170, 2021.
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- [44] Simeng Qiu, **Qiang Fu**, Congli Wang, and Wolfgang Heidrich. "Linear polarization demosaicking for monochrome and colour polarization focal plane arrays." *Comput. Graph. Forum.*, vol. 40, pp.77–89, 2021.
- [45] Qilin Sun, Congli Wang, **Qiang Fu**, Xiong Dun, and Wolfgang Heidrich. "End-to-end complex lens design with differentiable ray tracing." *ACM Trans. Graph.*, 40(4): 1–13, 2021.
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Honors and Awards

- Excellent Student Award of Chinese Academy of Sciences, 2009.
- Excellent Student Award of Zutong Scholarship in Xi'an Institute of Optics and Precision Machinery, 2008.
- First Prize of the Western Area Student Scholarship of Chinese Academy of Sciences, 2008.
- Excellent Graduate of University of Science and Technology of China, 2007.
- First Prize of Zutong Scholarship in University of Science and Technology of China, 2007.
- Second-class Scholarship in University of Science and Technology of China, 2004, 2005, 2006.